

Kubernetes Exercises

Name: Srun Nai Eang

ID: P20230006

1. Check number of pods in default namespace

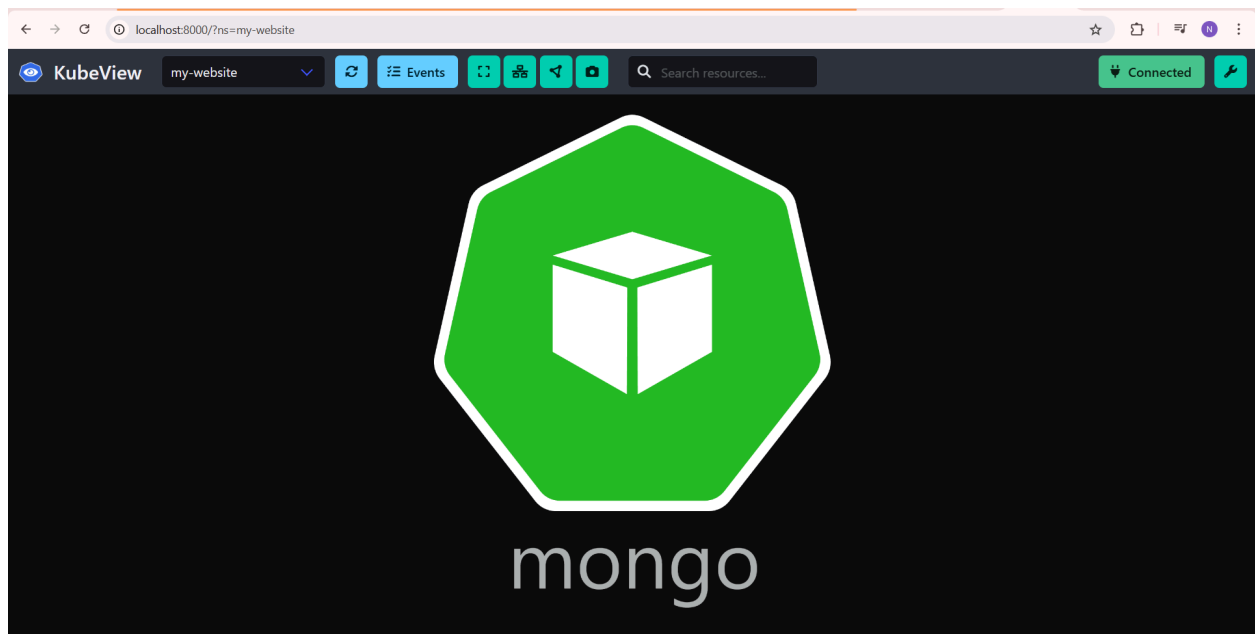
```
$ kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
custom-nginx	1/1	Running	0	9m26s
kubeview	1/1	Running	0	19m
my-app-59854d5646-kbrss	1/1	Running	0	21m

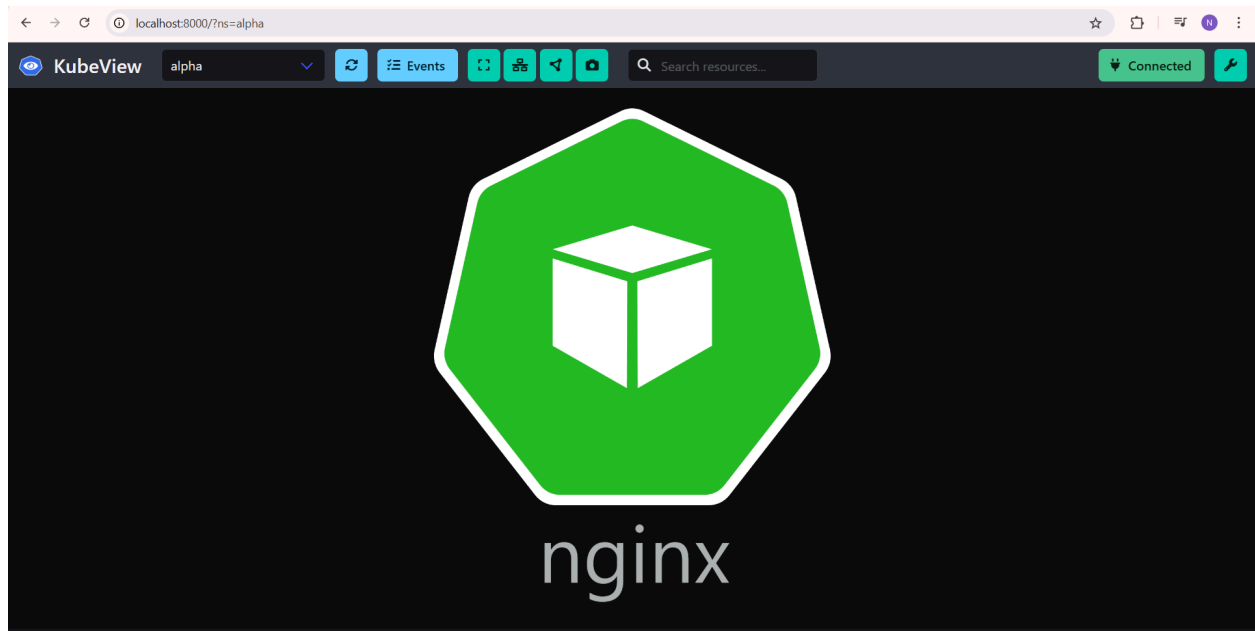
2. Create a new pod with name `nginx` and using the `nginx` image

```
$ kubectl run nginx --image=nginx  
pod/nginx created
```

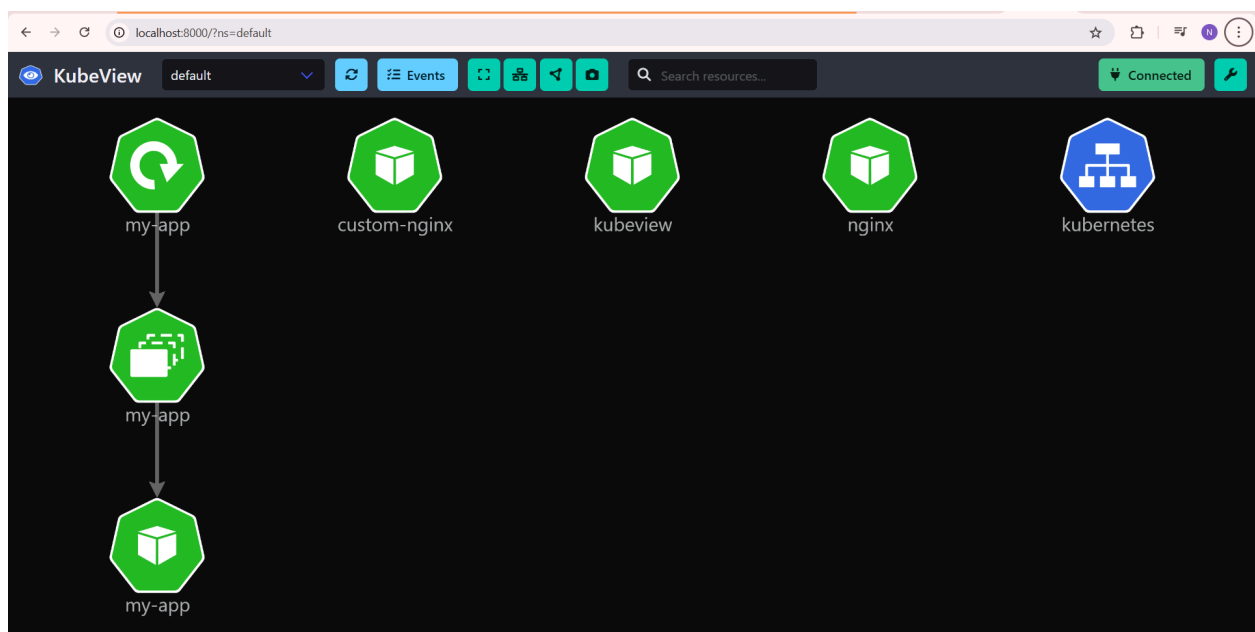
3. Create a pod named `mongo` using image `mongo` in a new Kubernetes namespace `my-website`



4. Create a new pod with name `nginx` and using the `nginx` image in the `alpha` namespace



5. Create a new pod `custom-nginx` using the `nginx` image and expose it on container port 8080.



6. Check which node the pod is hosted on

```
$ kubectl get pods -o wide
```

NAME	READY	STATUS	RESTARTS	AGE	IP	NODE	NOMINATED NODE	READINESS GATES
custom-nginx	1/1	Running	1 (106m ago)	9h	10.1.0.29	docker-desktop	<none>	<none>
kubeview	1/1	Running	1 (106m ago)	9h	10.1.0.27	docker-desktop	<none>	<none>
my-app-59854d5646-kbrss	1/1	Running	1 (106m ago)	9h	10.1.0.25	docker-desktop	<none>	<none>
nginx	1/1	Running	0	6m3s	10.1.0.30	docker-desktop	<none>	<none>

7. Get only the pods name

```
$ kubectl get pods -o name  
pod/custom-nginx  
pod/kubeview  
pod/my-app-59854d5646-kbrss  
pod/nginx
```

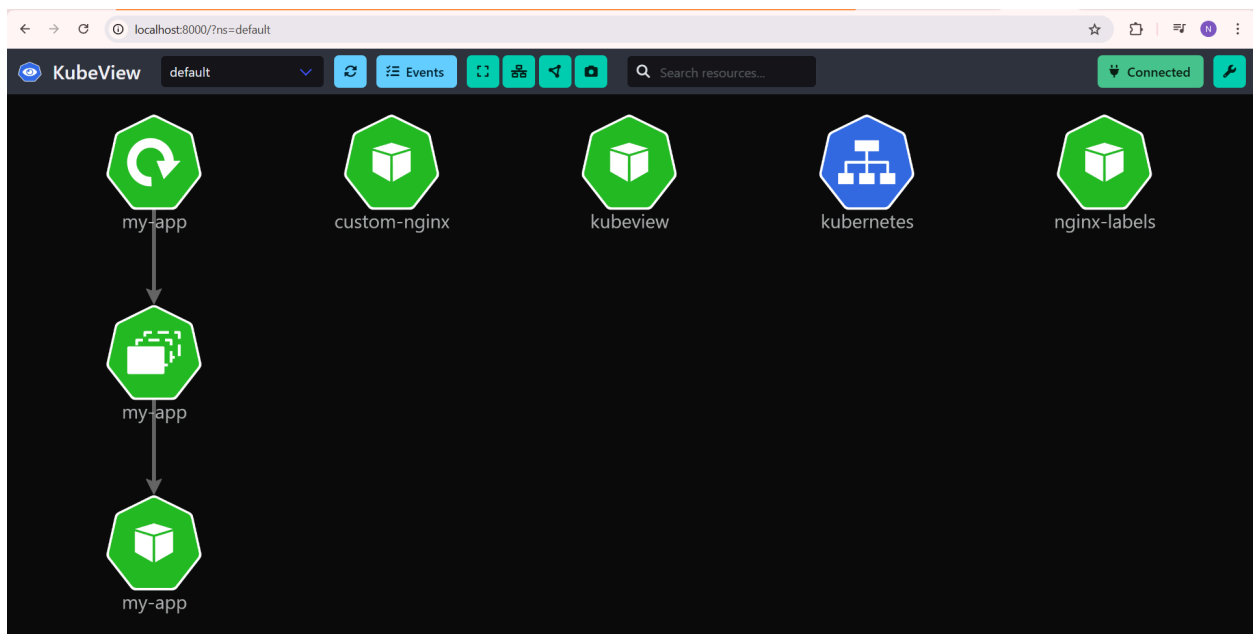
8. Delete the pod with the name nginx

```
$ kubectl delete pod nginx  
pod "nginx" deleted from default namespace
```

9. Delete the pod with the name nginx in the `alpha` namespace

```
$ kubectl delete pod nginx --namespace=alpha  
pod "nginx" deleted from alpha namespace
```

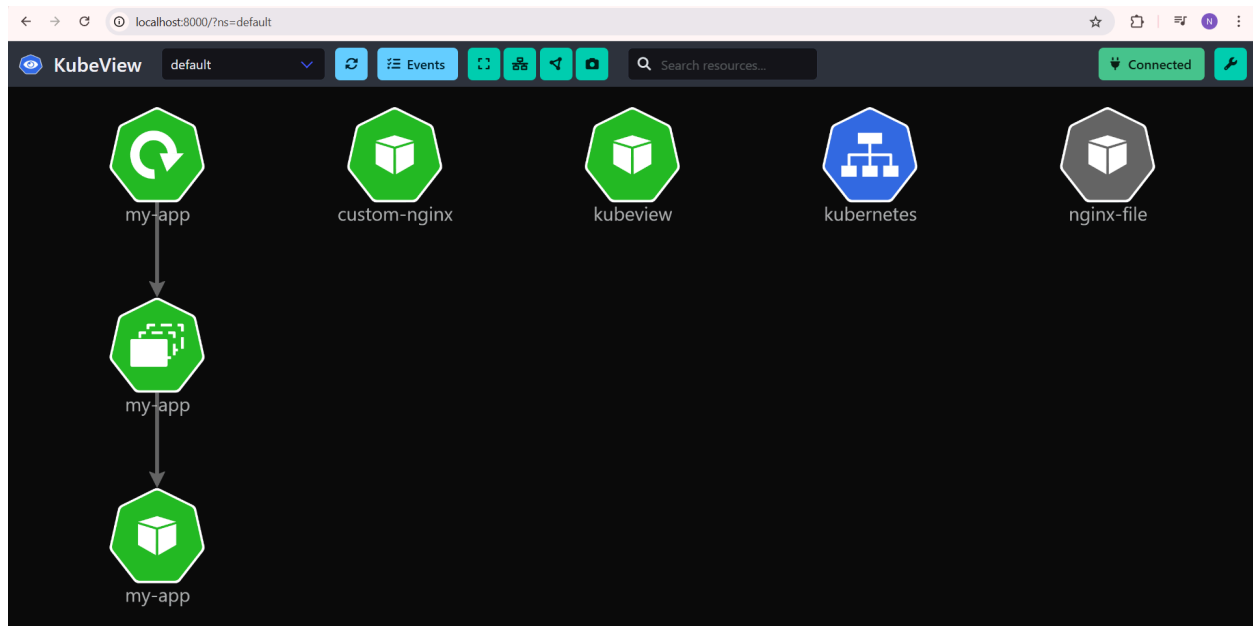
10. Create pod `nginx-labels` with `nginx` image and label `name=nginx`, `tier=frontend`, `env=dev`



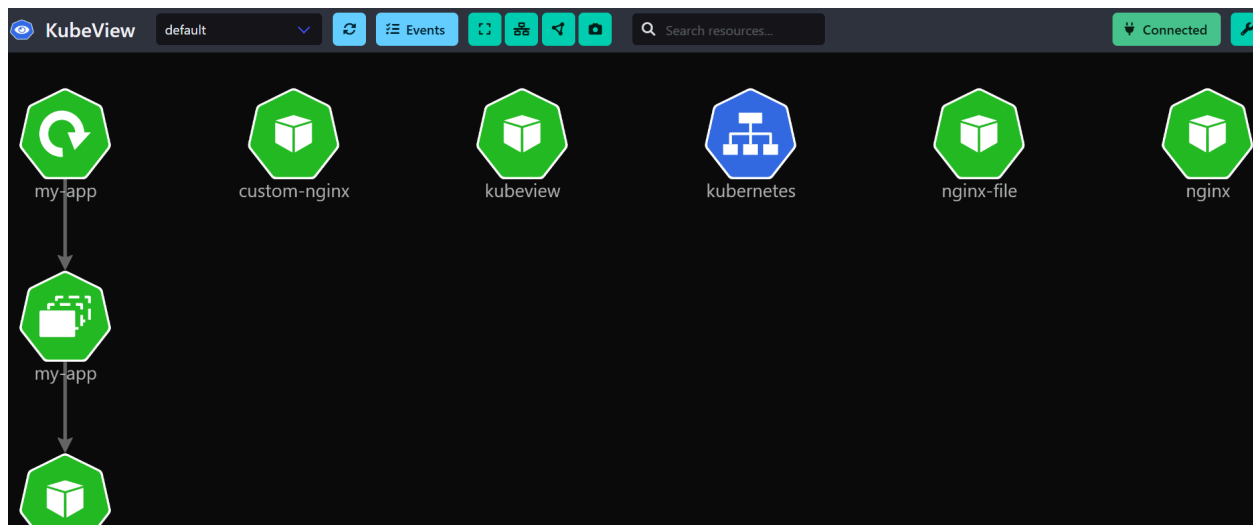
11. Delete the pod with name `nginx-labels` with force and no grace period

```
$ kubectl delete pod nginx-labels --force --grace-period=0
Warning: Immediate deletion does not wait for confirmation that the running resource has been terminated. The resource may continue to run on the cluster indefinitely.
pod "nginx-labels" force deleted from default namespace
```

12. Create a pod with name `nginx-file` and image `nginx` using pod definition file



13. Create a nginx pod with name `nginx` and copy the pod definition file to a `nginx_definition.yaml` file



14. Create a `ubuntu-1` pod with image `ubuntu` with command `sleep 4800`

